

Sean (Sieon) Kim

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Education

Dartmouth College

Bachelor of Arts in Mathematics

Hanover, NH

Graduation expected 2029

- **GPA:** 3.93/4.0
- **Activities:** First Year Research in Engineering Experience (FYREE), Grader for Math 35 (Real Analysis)
- **Coursework:** Metric Spaces & Measure Theory*, Algebra (Honors), Differential Topology*, Field & Galois Theory*, Commutative Algebra*, Complex Analysis*, Algebraic Topology*, Vector Bundles & Characteristic Classes**, Linear & Multilinear Algebra**, DRPs in C^* -algebras & K-theory, Category Theory, Algebraic Geometry, Algebraic Number Theory**

* denotes graduate/cross-listed graduate coursework, ** denotes planned coursework

Publications and Expository Works

- [1] *A note on the periodic Hilbert transform on a strip*, with Javier Gómez-Serrano; Journal of Differential Equations 422 (2025) 597-603. <https://doi.org/10.1016/j.jde.2025.01.022>
- [2] *Currents in Geometric Measure Theory*, (2023). Expository paper on White's compactness theorem for integral currents and its application to Plateau's problem | [Link to paper](#)

Research Experience

Computing Knot Floer Homology of (1,1)-Unknot Patterns

Advised by Abhishek Mallick (Dartmouth) and Ina Petkova (Dartmouth), with the SHUR REU

Jan. 2026 – Jun. 2026

Hanover, NH

- Ongoing research in low-dimensional topology

Manifold Learning for Low-Dimensional Data Classification

Advised by Peter Chin (Dartmouth), affiliated with the LISP Lab

Jan. 2026 – Jun. 2026

Hanover, NH

- Implementing manifold approximation using GMRA algorithms for feature extraction on high-dimensional data

Selective Spectrum Smoothing for Ensemble-based Data Assimilation

Advised by Yoonsang Lee (Dartmouth) and Bosu Choi (GSU)

Sep. 2025 – Jan. 2026

Hanover, NH

- Benchmarking spectrum smoothing methods for the ensemble transform Kalman filter across turbulence regimes

Bounding the Hilbert Transform Kernel on a Strip

Remotely advised by Javier Gómez-Serrano (Brown)

Jul. 2024 – Nov. 2024

Seoul, South Korea

- Proved a conjecture by Constantin-Strauss-Vărvăruță related to the finite depth water wave problem, published [1]
- Explored Jacobi theta identities with Mathematica to provide a closed form for the Hilbert transform kernel

Optimization of a Bullet

Independent | [Link to paper](#)

Sep. 2022 – Nov. 2023

Seoul, South Korea

- Modeled the air-resistance minimizing curve generating a radially symmetric bullet with the calculus of variations
- Analytically evaluated a formula for the curve when restricted to the quadratic case

Conferences and Workshops

Canadian Applied and Industrial Mathematics Society Conference

Poster presentation: *Optimization of the Aerodynamic Qualities of a Bullet* | [Link to poster](#)

Jul. 2024

Queen's Mathematics Summer School

Workshops on scalar conservation laws, machine learning theory, & Riemann surfaces (accepted w/ funding)

Jun. 2024

Awards and Honors

4x Citations for Academic Excellence — Dartmouth College

Outstanding performance in Math 71, 81, 104, 121; 2.4% of student grades are citations

Dec. 2025

Jack Byrne Scholarship — Dartmouth College

Grant totaling \$30,000 for enrichment activities in mathematics awarded to 9 students in the class of 2029

Mar. 2025